



Subject Code: BMC205

[illegible]

(SEM. II) THEORY EXAMINATION 2025-26

DATA STRUCTURES & ANALYSIS OF ALGORITHMS

M.MARKS: 70

TIME: 3 HRS

Note: Attempt all Sections. In case of any missing data; choose suitably.

$$02 \times 7 = 14$$

1. Attempt *all* questions in brief.

Q no.	Question	CO	Level
a.	Define data structure and its types.	1	K2
b.	What is asymptotic notation?	1	K2
c.	Define sparse matrix.	1	K2
d.	What is stack ADT?	2	K2
e.	Define binary search.	2	K2
f.	What is graph?	3	K2
g.	Define AVL tree.	4	K2

$$07 \times 3 = 21$$

2. Attempt any *three* of the following:

Q no.	Question	CO	Level
a.	Explain algorithm properties and performance analysis with asymptotic notations.	1	K3
b.	Explain singly and doubly linked lists with operations.	1	K3
c.	Explain stack applications: infix to postfix conversion and evaluation.	2	K3
d.	Explain sorting algorithms: selection sort and bubble sort with comparison.	3	K3
e.	Explain divide and conquer and greedy approach with suitable examples.	5	K3

$$07 \times 1 = 07$$

3. Attempt any *one* part of the following:

Q no.	Question	CO	Level
a.	Derive index formula for 2D array in column-major order with example.	1	K4
b.	Explain circular linked list and its operations with diagram.	1	K4

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question	CO	Level
a.	Explain recursion with Fibonacci series and Tower of Hanoi.	2	K4
b.	Explain queue operations and circular queue implementation.	2	K4



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5. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain heap sort and counting sort algorithms.	3	K4
b.	Explain adjacency list and adjacency matrix representation of graphs.	3	K4

6. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain binary search tree operations (insertion and deletion).	4	K4
b.	Explain threaded binary trees and Huffman coding.	4	K4

7. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain Strassen's matrix multiplication algorithm.	5	K4
b.	Explain Dijkstra and Bellman-Ford algorithms for shortest path.	5	K4

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